Panav Gohil

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PROFILE

First-year B.Tech student in Electronics Engineering (VLSI Design & Technology) at Delhi Technological University (DTU) with interests in Artificial Intelligence, Computer Vision, Embedded Systems, and Intelligent Hardware Systems. Experienced in developing hands-on projects involving machine learning, gesture recognition, driver monitoring, and autonomous sensing platforms. Passionate about building systems that combine software intelligence with real-world hardware to solve practical problems. Actively exploring open-source development, emerging technologies, and research opportunities while strengthening engineering fundamentals.

EDUCATION

Delhi Technological University (DTU) | New Delhi, India

Bachelor of Technology (B.Tech) | Expected Graduation: 2029

- **Current CGPA:** 7.5
- **Academic Achievements:** Secured 98.54 Percentile in JEE Main
- **Previous Education:** Standard XII: 92% | Standard X: 93.6%

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Google Apps Script
- **Artificial Intelligence & Machine Learning:** Machine Learning, Scikit-learn, Feature Engineering, Model Evaluation
- **Computer Vision:** OpenCV, MediaPipe, Image Processing, Real-Time Vision Systems & Gesture Recognition
- **Embedded Systems & Hardware:** Arduino, ESP32-CAM, Sensor Integration, IoT Prototyping, Hardware Interfacing
- **Tools & Technologies:** Git, GitHub, VS Code
- **Core Concepts:** Data Structures & Algorithms, Problem Solving, Object-Oriented Programming
- **Professional Skills:** Technical Research, Documentation, Public Speaking, Team Collaboration

OPEN SOURCE EXPERIENCE

GirlScript Summer of Code (GSSOC) 2026 | *Participant – AI/Agents Track*

- Selected to collaborate with a global developer community in a highly competitive open-source program.
- Exploring modern AI architectures through the AI/Agents track, expanding my knowledge from foundational ML into generative models and practical implementations.
- Gaining hands-on exposure to collaborative software development, version control, and open-source contribution workflows.

PROJECTS

Autonomous SENTRY Robot: Reconnaissance & Hazard Detection Platform | *Arduino, ESP32-CAM, IoT Sensors* | *April 2026 – May 2026*

- Engineered an autonomous robotics system designed for real-time reconnaissance and environmental hazard detection.
- Integrated ultrasonic and PIR sensors for navigation and motion detection, alongside MQ2 gas and flame sensors for physical and chemical threat identification.
- Programmed a servo-mounted laser system and automated buzzer alerts to provide immediate feedback upon hazard detection, bridging hardware logic with practical safety applications.

Driver Drowsiness Detection Using CNN | *Python, TensorFlow, OpenCV, CNN* | June 2026

- Developed a real-time driver drowsiness detection system capable of monitoring eye states through live webcam input to identify signs of fatigue.
- Built and trained a Convolutional Neural Network (CNN) to classify eye conditions as Open or Closed using a custom image dataset.
- Integrated OpenCV-based face and eye detection with the trained deep learning model to perform real-time inference and continuous driver monitoring.
- Implemented an alert mechanism that triggers warnings when prolonged eye closure is detected, improving safety by reducing the risk of fatigue-related accidents.
- Designed and deployed an end-to-end pipeline covering data preprocessing, model training, evaluation, and real-time prediction.

Real-Time Hand Gesture Recognition | *Python, OpenCV, MediaPipe, Scikit-learn* | *June 2026*

- Developed a real-time hand gesture recognition system using MediaPipe and OpenCV to detect and track 21 hand landmarks from live webcam input.
- Built a custom dataset and engineered 42 relative-coordinate features for robust gesture classification independent of hand position.

- Trained and evaluated a Random Forest classifier achieving 98.75% classification accuracy across gestures including Open Palm, Fist, Point, and Thumbs Up.
- Implemented an end-to-end machine learning pipeline covering data collection, feature extraction, model training, evaluation, and real-time prediction with confidence scoring.

Gesture Controlled System | *OpenCV, MediaPipe, Python* | *May 2026 – June 2026*

- Built a real-time gesture recognition system that allows users to control a computer without traditional input devices.
- Utilized OpenCV and MediaPipe to track hand landmarks and accurately identify gestures for system interaction.
- Implemented functionalities including cursor navigation, clicking, dragging, and scrolling to improve accessibility and user experience.

Stateline | *Google Apps Script, Custom Frontend* | *Jan 2026 – Feb 2026*

- Built an attendance tracking application to manage user schedules, branch data, and detailed attendance logs.
- Implemented features for manual data entry and comprehensive data export capabilities.
- Utilized Google Apps Script to establish a reliable backend system connecting user inputs with structured data management.

Vantage Finance | *Google Apps Script, Custom Frontend* | *Nov 2025 – Dec 2025*

- Developed a comprehensive financial management platform featuring a personal dashboard for budget limits, expense tracking, and loan management.
- Engineered backend functionality using Google Apps Script to handle payment mode tracking and automated invoice generation.

ACHIEVEMENTS

- Secured 98.53 Percentile in Joint Entrance Examination (Main), achieving an All India Rank (AIR) of 22,540 among 1.5+ million applicants.
- Awarded 1st Position in a university-level Ideathon for proposing and presenting an innovative technology-based solution.
- Represented Delhi Technological University as a Speaker at the 19th IIT Bombay Debate.
- Represented Delhi Technological University as a Speaker at the IGDTUW Parliamentary Debate 2026.
- Participated as a Speaker at the IIT Delhi Asian Parliamentary Debate 2026.
- Selected as a Participant in GirlScript Summer of Code (GSSOC) 2026 – AI/Agents Track.